

Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	3 Marks

High-Level Architecture



Module Architecture

`src/` - Core Business Logic

Module	File	Responsibility
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Module	File	Responsibility
Preprocessing	`cleaner.py`	HTML removal, normalization, stopwords
Preprocessing	`tokenizer.py`	Tokenization for BiLSTM path
Preprocessing	`encoder.py`	Label encoding, integer encoding
Preprocessing	`pipeline.py`	Orchestrates full preprocessing flow
BiLSTM	`architecture.py`	Keras model definition
BiLSTM	`trainer.py`	Training loop with callbacks
BiLSTM	`inference.py`	Load weights + predict
BERT	`architecture.py`	HuggingFace model definition
BERT	`trainer.py`	HuggingFace Trainer fine-tuning
BERT	`inference.py`	Tokenize + forward pass + predict
Detection	`rule_based.py`	Keyword dictionary + confidence boost
Detection	`ensemble.py`	Weighted average of model outputs
Detection	`mixed_emotion.py`	Gap threshold for mixed detection
Detection	`detector.py`	Unified detection interface (singleton)
Detection	`gemini_detector.py`	Cloud-mode: Gemini as emotion classifier
Gemini	`client.py`	API client with retry + backoff
Gemini	`prompt_builder.py`	Emotion-specific prompt templates
Gemini	`guidance_generator.py`	Orchestrates Gemini call + formatting
Analytics	`logger.py`	CSV interaction logging
Analytics	`aggregator.py`	KPI computation from logs
Analytics	`charts.py`	5 Plotly chart generators

Deployment Architecture

Environment	Mode	Models Used	Entry Point
Local (Full)	Full ML Pipeline	BiLSTM + BERT + Gemini	`app/main.py`
Streamlit Cloud	Demo Mode	Gemini API only	`streamlit_app.py`

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Student Input
  Text Preprocessing (NLTK)
  BiLSTM Inference (TF/Keras)
                                Weighted Ensemble
  BERT Inference (PyTorch)
  Rule-Based Keyword Boost
  Mixed Emotion Check
  Final Emotion + Confidence
  Gemini Prompt Builder
  Google Gemini API Call
  5-Section Guidance Response
  CSV Logger
  Streamlit UI Render
```

Config & Settings

File	Purpose
`config/settings.py`	All hyperparameters, paths, thresholds
`config/emotions.py`	Emotion labels, keywords, colors, metadata
`.env`	GEMINI_API_KEY (local)
`.streamlit/secrets.toml`	GEMINI_API_KEY (Streamlit Cloud)